11TH-STANDARD SCIENCE & INNOVATION PROJECT STUDY SHEET

AI Haptic Vision Navigation Headband

"See the Path. Feel the Direction. Navigate with Confidence."

Proof of Concept (POC): Currently runs locally via laptop webcam & pretrained YOLOv8. ESP32-CAM headband & physical wristband are planned for future hardware.

1. WHAT IS THE PROJECT?

An assistive visual guidance system that acts as an **electronic guide** for visually impaired users. It detects obstacles in front of the user and provides instant directional navigation through **spoken voice** and **wrist vibrations**.

2. WHAT PROBLEM DOES IT SOLVE?

The traditional white cane detects ground-level objects, while this system provides **complementary assistance for upper-body & head-level obstacles** (such as hanging branches, open truck doors, or raised signboards) to improve walking confidence.

3. HOW IT WORKS (SIMPLE 6-STEP WORKFLOW)

1. **Camera** — Captures video
2. **OpenCV** — Reads frames
3. **YOLOv8** — Identifies objects
4. **Position** — Left/Center/Right
5. **Navigation** — Decides safe move
6. **Guidance** — Voice + Vibration

4. HORIZONTAL POSITION ZONES

The camera view is split into three horizontal zones:

LEFT	CENTER	RIGHT
0% – 33%	33% – 66%	66% – 100%

The system finds which zone an obstacle occupies and calculates approximate visual closeness based on bounding-box size.

5. SIMPLE NAVIGATION RULES

OBSTACLE DETECTED	DIRECTION	GUIDANCE ACTION
Obstacle on **LEFT**	→	**MOVE RIGHT**
Obstacle on **RIGHT**	→	**MOVE LEFT**
Obstacle in **CENTER**	→	**STOP / SLOW DOWN**
Path is **CLEAR**	→	**MOVE STRAIGHT**

6. CORE TECHNOLOGIES (1 SENTENCE EACH)

- **Python:** The programming language used to connect the camera, AI detection, navigation logic, and audio output together.
- **OpenCV:** A computer vision library that captures and processes live video frames from the webcam in real time.
- **YOLOv8:** A fast, pretrained AI model that instantly recognizes common everyday objects like people, chairs, cars, and bicycles.

7. DUAL SENSORY FEEDBACK

Spoken Voice
Speaks clear directions like *"Move right"* or *"Stop"* through audio.

Wrist Vibration
Gives tactile directional pulses on the wrist when ambient noise is loud.

Why both? Dual feedback ensures the user still receives directional alerts even if traffic or crowd noise drowns out the sound.

8. CURRENT POC VS. FUTURE HARDWARE

MODULE	CURRENT LAPTOP POC	FUTURE HARDWARE
Camera	Laptop Built-in Webcam	ESP32-CAM on Headband
AI Engine	Laptop CPU (Pretrained YOLOv8)	Wearable Micro-Compute Unit
Voice	Laptop Speakers (MP3 Audio)	Bluetooth Earbud
Haptics	Dashboard Screen Simulation	Physical Vibration Wristband

9. KEY PROJECT STRENGTHS

- **Hands-Free Operation:** Worn naturally without carrying handheld scanners.
- **100% Offline & Private:** Runs locally on the machine with zero cloud dependency.
- **Rapid Response:** Computes navigation directions in fractions of a second.
- **Assistive Tool:** Designed to assist—not replace—traditional mobility tools.

5 THINGS TO REMEMBER

1. Camera sees
2. YOLO detects
3. Position is found
4. Navigation decides
5. Voice + vibration guide